

Are hatchery-reared abalone naïve of predators? Comparing the behaviours of wild and hatchery- reared northern abalone, *Haliotis kamtschatkana* (Jonas, 1845)

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Abstract

Abalone populations have declined worldwide, generating interest in enhancement using hatchery-reared individuals. In many cases, such restoration efforts have met with limited success due to high predator-induced mortality rates. Furthermore, the mortality rates of outplanted hatchery abalone are often considerably higher than for wild individuals. This study uses northern abalone (*Haliotis kamtschatkana*) as a case study to determine whether hatchery-reared abalone behave differently than their wild counterparts. In the field, outplanted hatchery-reared abalone were significantly less responsive than wild abalone, in terms of number of abalone responding and intensity of response, to nearby movement and to physical contact with an inert probe. Also, when encountering a cue to which all abalone responded (a seastar predator), hatchery-reared individuals remained subdued. Anti-predator behavioural deficits in hatchery-reared abalone were more pronounced in 4-year-old individuals than in 1-year-old individuals, suggesting an influence of either age or amount of time spent in the hatchery environment. These behavioural differences are expected to increase the vulnerability of hatchery-reared abalone to predators, and are likely a major cause of their elevated predator-induced mortality when outplanted.

Keywords: conservation, marine, benthos, invertebrates, molluscs, *Haliotis kamtschatkana*

Introduction

Abalone populations have declined worldwide, in many cases as a direct result of overfishing (Sloan & Breen 1988; Karpov, Haaker, Taniguchi & Rogers-Bennett 2000; Hobday, Tegner & Haaker 2001; Shepherd, Rodda & Vargas 2001). There is accordingly widespread interest in stock enhancement using hatchery-reared individuals, both to rebuild collapsed populations and supplement those which are still subject to fishing pressures (McCormick, Herbinson, Mill & Altick 1994; Dixon, Day, Huchette & Shepherd 2006). However, restoration efforts have largely been limited by high (35–100%) and unpredictable post-outplanting mortality rates (reviewed in McCormick *et al.* 1994). The most intriguing aspect of outplant mortality is that the mortality rate of recently outplanted abalone is often considerably higher than the mortality rates reported for wild individuals of the same species, often in the same habitats (*Haliotis iris* (Gmelin) in New Zealand, Schiel 1993; *H. rufescens* (Swainson) in California, Rogers-Bennett & Pearse 1998; *H. midae* (Linnaeus) in South Africa, Sweijd, Snethlage, Harvey & Cook 1998; *H. laevigata* (Donovan) in Australia, Dixon *et al.* 2006; *H. kamtschatkana* (Jonas) in Canada, Hansen 2011; Hansen & Gosselin 2013; but see Schiel 1992 – *H. iris* in New Zealand). These trends all hold true for northern abalone, *H. kamtschatkana*, which accordingly presents a good case study for examining the limitations of outplanting and how they can be overcome.

Haliotis kamtschatkana is the only abalone species to be found in Canada. This species is distributed from Alaska to Baja California (Sloan & Breen 1988), and its populations in British Columbia, Canada, are endangered and failing to recover despite fishery closures. In fact, the northern abalone was the first marine invertebrate to be designated as threatened by the Committee on the Status of Endangered Wildlife in Canada. The abundance of *H. kamtschatkana* at index sites throughout British Columbia declined by 80% before the fishery was closed in 1990 (Hankewich & Lessard 2008); since then, population sizes have failed to increase despite the moratorium (Hankewich & Lessard 2008). This suggests restoration efforts are needed if abalone populations are to recover. One potential solution, the outplanting of hatchery-reared *H. kamtschatkana*, has so far been limited by high mortality rates of hatchery abalone when released into the wild (Read, Lemay, Acheson & Boulding 2012; Hansen & Gosselin 2013; Read, Lessard & Boulding 2013). For *H. kamtschatkana*, the mortality rate of recently outplanted juveniles has been estimated at 66% over 2 weeks (Hansen & Gosselin 2013), while the mortality rate of wild individuals of the same age in the same region is <1% over the same duration (Breen 1980, 1986; Fournier & Breen 1983; Sloan & Breen 1988). Given that predation is one of the major sources of mortality for outplanted abalone (Tegner & Butler 1985; Schiel & Welden 1987) and has been shown to be responsible for almost all outplant mortality in *H. kamtschatkana* (Hansen & Gosselin 2013), this raises the question as to why hatchery-reared abalone are more vulnerable to predators than their wild counterparts. Answering this question is important if outplanting is to become an effective restoration strategy.

Juvenile northern abalone are preyed upon by a variety of predators, including *Scyra acutifrons* (Dana), *Lophopanopeus bellus* (Stimpson), *Cancer productus* (Randall), *Pycnopodia helianthoides* (Brandt), *Enteroctopus dofleini* (Wülker), *Enhydra lutris* (Linnaeus), *Scorpaenichthys marmoratus* (Ayres) and *Anarrhichthys ocellatus* (Ayres) (Sloan & Breen 1988; Griffiths & Gosselin 2008). The ability of hatchery-reared abalone to avoid predation is accordingly critical to their survival upon outplanting. The influence of outplant density and habitat features on subsequent outplant survival have been the focus of numerous studies (McCormick *et al.* 1994; Shepherd, Preece & White 2000;

Shepherd & Clarkson 2001; deWaal & Cook 2001; Dixon *et al.* 2006; Roberts, Keys, Prendeville & Pilditch 2007; Read *et al.* 2013), and can therefore be regulated through careful selection of outplanting methods and sites. In comparison, behavioural anomalies among hatchery-reared abalone are relatively unstudied (Schiel & Welden 1987; Strauss & Friedman 2009).

Rearing organisms in a hatchery environment often leads to behavioural deficits due to lack of ecologically relevant experience and/or inadvertent selection for maladaptive phenotypes (Jackson & Brown 2011). For example, ecologically relevant experiences lacking in the hatchery environment include exposure to structurally complex habitats, predator cues and physical contact with predators. In the absence of these experiences, learnt predator recognition and avoidance behaviours may not be adequately expressed (Olla, Davis & Ryer 1998; Álvarez & Nicieza 2003; Jackson & Brown 2011). Hatchery conditions can accordingly result in a lack of proper behaviour or the development of new behaviours which make organisms more vulnerable to predators (McCormick *et al.* 1994). A seminal paper by Schiel and Welden (1987) demonstrated that hatchery-reared red abalone (*H. rufescens*) behave differently than wild individuals in the presence of predators in a laboratory environment. It has since been postulated that behavioural differences may be responsible for the increased vulnerability of hatchery-reared abalone of various species to predators upon outplanting (McCormick *et al.* 1994). The above studies, however, were carried out in a laboratory setting, and involved wild abalone that had been collected in the field and transferred to the laboratory. It is not clear whether such behavioural differences occur in the field, after hatchery-reared abalone have been outplanted.

The purpose of this study was therefore to determine if outplanted hatchery-reared *H. kamtschatkana* behave differently from undisturbed wild individuals in the field, in a way that would leave hatchery-reared abalone more vulnerable to predators than wild individuals. The specific objectives of this study were to determine (i) whether wild and hatchery-reared northern abalone in the field respond differently to shadows, movement, mechanical stimulation or contact with one of their main predators, the sunflower seastar (*P. helianthoides*), and (ii) whether responses to predator cues vary among hatchery abalone of different age groups.

Methods

The hatchery-reared abalone used in this study were obtained from the Bamfield Huu-ay-aht Community Abalone Project in Bamfield, on the west coast of Vancouver Island, British Columbia, Canada. Abalone produced by this aquaculture operation originated from wild abalone broodstock collected nearby in Barkley Sound. All abalone used in these experiments were the F1 stock from wild caught adults. The field experiment described herein was also conducted in Barkley Sound. Laboratory experiments were conducted at the Bamfield Marine Sciences Centre.

Responses of wild and hatchery-reared abalone

To determine if outplanted hatchery-reared *H. kamtschatica* exhibit behaviours that differ from their wild counterparts, we carried out a field experiment examining the responses of wild and hatchery-reared abalone to shadows, contact with an inert object, and contact with a major predator. The experiment took place at a site on the north-west side of Fleming Island (48°52'701"N, 125°09'698"W) at which we had previously outplanted a total of 820 hatchery-reared northern abalone (4.23–6.26 cm SL, approximately 4 years old) on two dates (July 2nd and August 1st), 3–7 weeks prior to the start of the present experiment. The mean shell length of hatchery abalone at the time of the behavioural experiment was 5.22 ± 0.08 cm (SE). The outplanting site was chosen for several features, including (i) the lowest combined predator density (0.079 ± 0.045 individuals/m²) of five potential sites that were surveyed; (ii) the presence of structurally complex habitat (cobble, boulders and bedrock with crevices); and (iii) the presence of wild abalone (these were at a density of 0.033 ± 0.016 individuals/m²) (see Hansen & Gosselin 2013 for more details). This experiment involved trials carried out on five occasions over a 2-week period, in which all abalone found by SCUBA divers were individually exposed to a given stimulus and the subsequent response of the abalone recorded. The majority of hatchery-reared individuals were identified by tags, although some additional hatchery-reared abalone that had lost their tags were recognised by their readily distinguishable shell colour, shell scarring at the tag attachment site and lack of attached algae or sessile organisms.

Each abalone was haphazardly assigned one of three stimulus treatments; the stimulus was applied to the individual *in situ* and its response and initial position, whether cryptic or emergent, were recorded. We therefore recorded observations for a range of sizes of abalone and for both wild and hatchery-reared individuals. The three treatments were: (i) displacing rocks and hovering near the abalone ($n = 36$ wild and 27 outplanted abalone); (ii) touching the abalone with a plastic probe for 30 s ($n = 15$ wild and 20 outplanted abalone) and (iii) touching the abalone with a live *P. helianthoides* for 30 s ($n = 12$ wild and 12 outplanted abalone). Treatment (i) served as a control for the effects of the contact cues of treatments (ii) and (iii). The control also revealed the effect of habitat disturbance without directly contacting the abalone. We recorded the timing (i.e. how long after the application of the cue) and type of all reactions over a 3 min period following application of the treatment. Types of reactions observed were: extension of the cephalic tentacles, extension of the epipodium and associated tentacles, mushrooming, clamping of the shell, twisting of the shell, release of a viscous fluid and rate of movement, as described in Table 1 and Supporting Information. These abalone behaviours have all been previously described and are known to be associated with predator avoidance (Bullock 1953; Clark 1958; Feder 1963; Montgomery 1967). After each trial, the origin of every abalone (i.e. whether hatchery-reared or wild) was recorded.

The responses of abalone were scored on a numerical scale ranging from 0 (no visible reaction) to 8 (strongest reaction) (Table 2). This 8-point scale took into account response types but also response times, as these have both been shown to be good measures of the intensity of responses of marine invertebrates to predators (Legault & Himmelman 1993). The scale has discrete categories and very clear requirements to fit reactions into one category or another, such that there is no potential for observer bias in assigning a reaction grade to an abalone. For example, galloping would place an abalone in one of the 5–8 categories, while the time at which the galloping behaviour was initiated would determine precisely which of those categories the abalone would fall into. This scale effectively addressed all combinations encountered in this experiment. Statistical analyses were performed in SPSS version 19 and R 2.15.2. Multinomial logistic regression was used

Table 1 Description of northern abalone behaviours

Behaviour	Description	Illustration
Epipodium extended	The epipodium and associated tentacles are extended beyond the margin of the shell	Supporting Information <i>a</i>
Cephalic tentacles extended	Cephalic tentacles are extended and wavering	na
Mushrooming	The foot contracts into a columnar shape, raising the shell	Supporting Information <i>b</i>
Slow movement	The abalone moves slowly, the entire margin of the foot appears to remain in contact with the substrate	na
Fast movement	The abalone moves quickly, the entire margin of the foot appears to remain in contact with the substrate	na
Galloping	The abalone moves rapidly, waves of muscle contraction are apparent along the margin of the foot, and the abalone appears to be rocking	na
Clamping	The epipodium and cephalic tentacles are retracted and the shell is pulled down to the substrate	na
Twisting	The shell is lifted from the foot and twisted back and forth almost 180 degrees	Supporting Information <i>c</i>
Fluid	The abalone expels a viscous fluid from the respiratory pores	na
'Tail' over shell	The posterior end of the foot is raised onto the shell and swept around the perimeter of the shell	Supporting Information <i>d</i>

Table 2 Scale for grading northern abalone reactions

Assigned value	Behaviours
0	No reaction
1	Only epipodial tentacles extended
2	Only cephalic tentacles extended, or cephalic and epipodial tentacles extended
3	Slow movement after 30 s
4	Slow movement within the first 30 s, or delayed but rapid movement (not galloping)
5	Galloping or twisting after 60 s, or immediate clamping
6	Galloping or twisting between 30 and 60 s after the cue
7	Galloping or twisting between 10 and 30 s after the cue
8	Galloping or twisting within 10 s of the cue

to determine which factors were significantly influencing abalone behaviour, as the scaled reaction grades fit a multinomial distribution. In this analysis, reaction grade was the response variable and the predictor variables were cue treatment (fixed; three treatments), type of abalone (fixed; two types of abalone: late-stage outplanted hatchery juveniles and wild abalone), and position (fixed; two positions: cryptic and emergent). Models were fit in a reverse step-wise fashion and the model with the lowest corrected Akaike's Information Criterion (AIC_c) was selected.

Effect of abalone age on response to predators

To determine if the age of hatchery-reared abalone influences the expression of predator avoidance behaviours, the responses of hatchery-reared abalone to nearby movements and to contact with either inert objects or a major predator were recorded in the laboratory using hatchery-reared northern abalone of two age classes: 1-year old, early-stage hatchery juveniles (0.94–2.16 cm shell length (SL)), and 4-year old, late-stage hatchery juveniles (4.41–5.93 cm SL). Abalone were placed in clear plastic containers (30 × 15 × 10 cm), one individual per container. Two sides of each container had been cut out and replaced with mesh to allow for water flow; the containers were then placed in a seawater table with flowing seawater, in which the water was maintained at a depth of 5 cm. Abalone were allowed to acclimatise to their container for 60 min prior to the start of the experiment. Although certain species of abalone require considerably longer acclimation periods (*H. discus hannai* (Ino) show greater movement up to a week following disturbance – Werner, Flothmann & Burnell 1995), 60 min was found to be sufficient in our pilot study. Treatment cues were first applied to the individual in the container that was the furthest downstream; once the response of this individual was recorded, it was removed from the water table and the next

individual, one container upstream, was exposed to the cue. In a pilot experiment, it was found that upstream abalone never reacted to downstream cues. Three treatments were applied: (i) control: removing the lid from the container without applying a subsequent cue ($n = 36$ early-stage juveniles and 32 late-stage juveniles); (ii) inert probe: removing the lid from the container, then touching the abalone with a clean plastic probe for 30 s ($n = 36$ early-stage juveniles and 45 late-stage juveniles) and (iii) predator: removing the lid from the container, then touching the abalone with *P. helianthoides* for 30 s ($n = 36$ early-stage juveniles and 42 late-stage juveniles). Treatments (ii) and (iii) were always applied to the posterior end of the abalone. In each treatment, we recorded the time and type of reactions of abalone during the 3 min following application of the cue.

The responses of abalone in the laboratory were scored on the same numerical scale used in the field experiment (Table 1). We also noted every case in which an abalone responded to a cue by raising the posterior end of its foot over the shell and sweeping it back and forth, as this was a previously undescribed response that might substantially increase vulnerability to predators. The analysis consisted of reverse step-wise multinomial logistic regressions with reaction grade as the response variable, and abalone age group (fixed; two treatments) and cue (fixed; three treatments) as the predictor variables. The multinomial logistic regressions were followed-up with a Wald chi-square test of the best model to compare among treatments.

Results

Responses of wild and hatchery-reared abalone

The responses of abalone, graded from 0 to 8 (no reaction to strong reaction), were found to be influenced by the type of abalone (wild or hatchery-reared), the position of the abalone (cryptic or emergent), and the cue applied (nearby movement, contact with an inert probe or contact with a major predator). Abalone reactions were best described by a model that included these three factors but no interaction terms. Notably, wild northern abalone responded more strongly to stimuli than their hatchery-reared counterparts ($\chi^2_1 = 17.541$, $P < 0.001$; Fig. 1a). In addition, cryptic abalone responded more strongly than

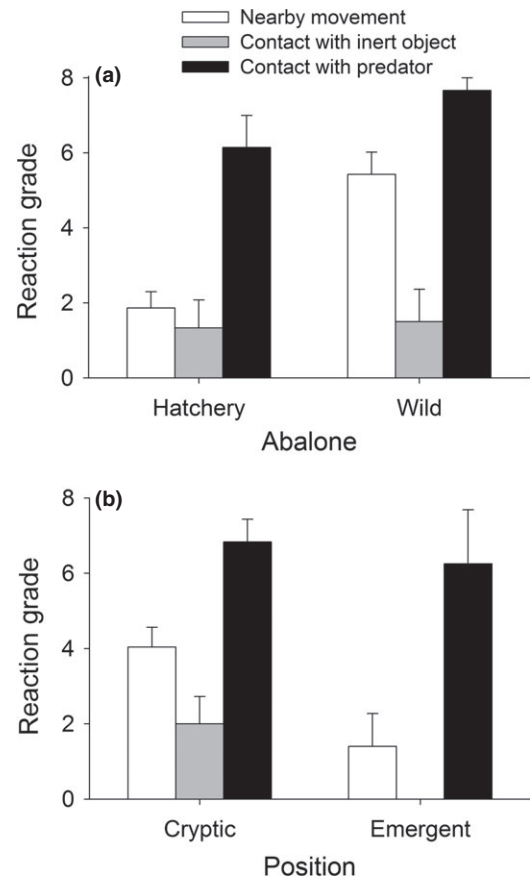


Figure 1 The reaction grades of abalone (a) of different origins, and (b) occupying different positions, in response to three cues. The cues are as follows: Control = Nearby movement; Contact with an inert object; Contact with a predator, *Pycnopodia helianthoides*. Error bars represent SE.

emergent ones ($\chi^2_1 = 8.263$, $P = 0.004$; Fig. 1b). For example, no emergent hatchery-reared juveniles responded to the control cue, whereas 60% of emergent wild abalone did respond to the control cue. As for cryptic wild abalone, almost 77% responded to the control cue by galloping, which is considered a strong reaction. This was only observed in 25% of hatchery-reared abalone. Similarly, emergent hatchery-reared abalone did not respond to contact with an unscented object, while 50% of emergent wild abalone did. Among the cryptic individuals, 33% of hatchery-reared abalone responded, 10% by clamping, and 77% of cryptic wild abalone responded, 17% by clamping. All abalone responded to contact with the seastar *Pycnopodia helianthoides* in a similar fashion. The sequence of reactions to this predator was

generally as follows: extension of the epipodial and cephalic tentacles either shortly before or in conjunction with immediate rapid (galloping) movement, with some individuals twisting just before or just after beginning to gallop. The response was slightly more rapid in wild individuals; galloping was initiated after 5.2 ± 0.9 s in wild individuals, and after 22.5 ± 13.6 s in hatchery-reared individuals (Wilcoxon Mann–Whitney Rank Sum Test: $Z = 1.977$, $P = 0.024$; Modified Levene's Test shows that the null hypothesis of homoscedasticity should not be rejected: $F_{1,18} = 1.205$, $P = 0.287$).

Since cue type was also found to be a significant predictor of reaction grade ($\chi^2_2 = 31.347$, $P < 0.001$), Wald Chi-Square *post hoc* tests were used to reveal where differences existed. Abalone responded more strongly to contact with predators than to either contact with an inert object or nearby movement, and responded more strongly to nearby movement than contact with an inert object (Fig. 1). An interaction between cue and position did not improve model performance ($AIC_c = 100.653$ as compared to an AIC_c of 99.341 for the best model), and was not significant ($\chi^2_2 = 4.751$, $P = 0.093$), yet all emergent hatchery-reared abalone failed to respond to the control and unscented prodding cues, while cryptic hatchery-reared individuals showed low to moderate responses. This suggests that the lack of significant interaction between cue and position may be due to small sample size.

Effect of abalone age on response to predators

The reactions of hatchery-reared abalone in the laboratory, graded from 0 to 8 (no reaction to strong reaction), were found to be influenced by an interaction between age group and cue type ($\chi^2_2 = 10.592$, $P = 0.005$) and correspondingly, the saturated model that included the interaction term had the lowest AIC_c score. This interaction resulted because early-stage hatchery juveniles responded more strongly to nearby movement and contact with an inert object than their late-stage counterparts, but both age groups responded similarly to contact with a major predator (Fig. 2).

In some cases, hatchery-reared abalone responded to stimulation with *P. helianthoides* by raising the posterior end of their foot and sweeping it over the surface of their shell (see Supporting Information). This behaviour was only ever observed in hatchery-reared individuals in the

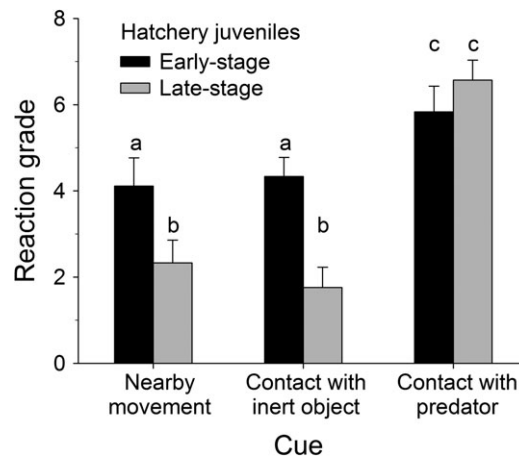


Figure 2 The reaction grades of early and late-stage hatchery-reared abalone in response to three cues when their initial position is either cryptic or emergent. The cues are as follows: Control; Unscented = prodding with an unscented probe; Pycnopodia = prodding with *Pycnopodia helianthoides*. Error bars represent SE. Values with the same letter are not significantly different, as determined by Wald chi-square tests.

laboratory environment. Four individuals from the early-stage juvenile size class ($n = 42$), representing 9.5% of this group, responded in this way, as did four individuals from the adult size class ($n = 36$), equivalent to 11.1%.

Discussion

The behaviours of hatchery-reared northern abalone differed from those of wild abalone, with hatchery abalone responses being significantly subdued compared to those of wild abalone. This was particularly evident in their responses to nearby movement and to being touched with an inert object. In fact, a far lower percentage of hatchery-reared abalone responded to these two cues than wild abalone and those that did respond were more apt to simply extend their tentacles or engage in slow movement, whereas wild individuals tended towards rapid movement (nearby movement and shadows cue) and clamping (inert object cue). This could be expected to make hatchery-reared abalone more vulnerable than wild abalone to predators; nearby movement and shadows might be associated with the approach of predators, and unscented physical contact could represent the approach of a predaceous crab (e.g. *Cancer productus*) from down-current, as these

predators are known to displace rocks to gain access to cryptic abalone (Schiel & Welden 1987). This hypothesis is supported by the clamping response of wild abalone to unscented contact, since clamping is an appropriate response to prevent dislodgement or predation by crabs (Cox 1962). Most hatchery-reared abalone responded to direct contact with *P. helianthoides* in the same manner as, but more slowly and with slightly lower intensity, than wild abalone. Once again, a delayed response to predator cues in hatchery-reared abalone would be expected to increase their vulnerability to *P. helianthoides* in particular, as the predator avoidance strategy of abalone for this seastar is to avoid contact with them, or to break physical contact and outpace them (Sloan & Breen 1988). Hence, wild and hatchery-reared abalone differed in their responses to all three cues, albeit to different degrees.

The behavioural responses of abalone were influenced by the initial position of the abalone. In general, the responses of cryptic abalone were stronger than those of emergent abalone, regardless of the cue that was applied. The absence of response to nearby movement or shadows among emergent hatchery-reared abalone was reflected to a lesser degree in wild abalone, with cryptic individuals responding more strongly to nearby movement than emergent ones. This difference in response may reflect that the cue itself was perceived differently by cryptic and emergent abalone. Notably, a cryptic abalone would be found when rocks above it were removed or the rock it was on was turned over, whereas emergent abalone were immediately visible. Thus, a cryptic abalone would experience an increase in light as it was exposed, in addition to the shadows and rock movement that constituted the cue. This explanation is supported by the fact that both hatchery-reared abalone and wild abalone responded more strongly to the control cue when their initial position was cryptic.

The observed behavioural differences between wild abalone and hatchery-reared abalone occurred despite the hatchery-reared abalone having been outplanted 1–2 months earlier and thus exposed to natural stimuli for some time prior to the behavioural experiment. This reveals that abalone do not re-learn wild-type behaviour when exposed to natural stimuli for 1–2 months. From this, we can deduce that there is no easy solution to the behaviour problem; abalone cannot be

trained to re-acquire appropriate predator avoidance or wild-type behaviours within a 1–2 month period. This conclusion is consistent with the findings of Hansen and Gosselin (2013), in which abalone were placed in field cages for 1 week, and thus protected from predators during that time; these abalone experienced high mortality as soon as they were released, comparable to mortality rates of abalone that had been outplanted directly onto the open substrate. It is not clear whether a long-term training approach would be more successful in reducing outplant mortality. However, Strauss and Friedman (2009) report that raising northern abalone in habitat-enriched tanks can prevent the development of inappropriate habitat selection behaviours. Future experiments should also examine whether exposing abalone to predators or predator cues throughout their development in the hatchery can reduce anti-predator behavioural deficits and improve post-outplanting survival.

When the behaviour of different age groups of hatchery-reared abalone was examined in the laboratory, late-stage hatchery juveniles were less responsive than early stage juveniles to being touched with an inert object. Late-stage hatchery juveniles were also less responsive to nearby movement than their early-stage counterparts. Although it is not clear whether these behavioural differences are due to ontogenetic changes or time spent in the hatchery environment, they are expected to be detrimental to the survival of abalone in the field. In the natural environment, the movement of nearby rocks can foreshadow the arrival of *Cancer productus*. Similarly, passing shadows could also indicate the approach of a motile predator. Hence, failure to respond correctly to these cues could increase vulnerability to predators for both early- and late-stage juveniles, since even large outplanted juvenile northern abalone (>4 cm SL) are vulnerable to predation by crabs, sea stars and octopus (Hansen & Gosselin 2013). The possibility that the reduced responsiveness of late-stage hatchery juveniles is due to habituation in the hatchery should be examined. For example, hatchery feeding practices, in which hatchery personnel regularly move along the tanks to remove deceased individuals and feed the abalone, might prompt hatchery abalone to learn to perceive nearby movement as a benign stimulus or even associate it with the arrival of food rather than as a potential threat. If this were the case,

hatchery-reared abalone would progressively become habituated to nearby disturbances and passing shadows, a behaviour that probably reduces stress and energetic costs and increases growth in a hatchery environment but is detrimental to their survival in the field.

Approximately 10% of hatchery-reared northern abalone responded to *P. helianthoides* by raising the posterior portion of the foot over the shell and sweeping it around the perimeter of the shell. This action is postulated to be maintenance behaviour, used to clear settling debris off of the shell (J. Lessard pers. comm.; J. Richards pers. comm.; pers. obs.). To our knowledge, this is the first report of abalone demonstrating such behaviour in response to a predator cue. The most similar abalone behaviour to have been reported is that of *H. rufescens* raising its mantle and extending its epipodium over its shell, whilst fleeing, in response to stimulation with *P. helianthoides* tube feet (Montgomery 1967). The latter serves as an effective escape behaviour by preventing *P. helianthoides* from firmly grasping the shell of the potential victim (Margolin 1964). However, in the case of the behaviour we observed in *H. kamtschatkana*, raising the posterior end of the foot would not free it from a predator because (1) that part of the foot can only cover a small portion of the shell (less than 20%, as opposed to the almost complete coverage of the shell by *H. rufescens*' mantle), and (2) the abalone cannot crawl or flee while raising the foot in this behaviour, whereas *H. rufescens* can gallop with its mantle raised. Since *P. helianthoides* consumes its prey by crawling over them and everting its stomach, an abalone that was not fleeing could be overtaken and killed. We did not observe the sweeping behaviour in wild abalone or in hatchery-reared abalone that had been outplanted into the wild for 4–8 weeks. Abalone exhibiting this behaviour in response to *P. helianthoides* could accordingly have rapidly fallen victim to predatory sea stars once outplanted, which accounts for the absence of this behaviour in the wild.

In conclusion, the behaviour of hatchery-reared abalone differs from that of wild abalone in important ways that relate to predator avoidance, such as responses to nearby movement and to physical contact. In addition, even when encountering a cue to which all abalone responded (i.e. contact with the predator *P. helianthoides*), the response of hatchery-reared abalone remained subdued compared to that of wild abalone. Hatchery-reared northern abalone also showed evidence of

exacerbated behavioural discrepancies among older individuals. Older hatchery abalone are significantly less responsive to shadows, nearby movement and contact with an inert object than younger individuals. All of these behavioural differences would increase the vulnerability of hatchery-reared northern abalone to predators, and are likely a major cause of the elevated predator-induced mortality of outplanted hatchery-reared abalone.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1: Northern abalone behaviours. (a) Extension of epipodium and associated tentacles, (b) mushrooming, (c) twisting, and (d) “tail” over shell.